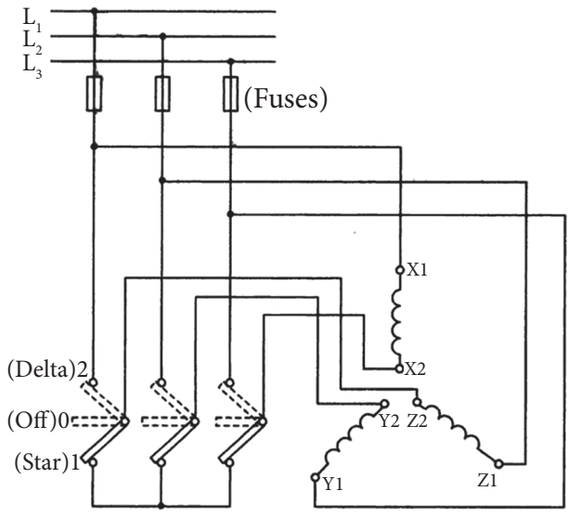



***Start-Delta Starter***

to 'start' the three motor windings are connected in star formation. Each phase windings gets only 58% of the line voltage. So the starting line current to the motor is reduced. As the motor picks up speed and attains nearly 80% of normal r.p.m, the starter handle is pushed to 'run' position. In this run position, the three winding get connected in delta formation so that each winding gets full supply voltage and full line current.

This starter is provided with usual protective devices such as overload relay, no volt coil release and single phase prevented. Star-delta starter is used for motors of capacity 5 hp to 15 hp.

## Auto Transformer Starter

An auto transformer is fitted in this starter and so it is costlier. Star connected transformers with 40, 60 and 80% tapings, allows specific voltage to be pressed against the motor windings.

When the starter handle is thrown to start position-reduced voltage is supplied to the motor windings and the motor starts with reduced line current and reduced torque. When the starter handle is pushed to run position, full line voltage is applied to motor windings and the auto transformer is isolated from the circuit. This starter is provided with usual protective devices such as overload relay and no volt coil. This type of starter is intended for starting three phase induction motors of capacity more than 15 hp.

## Rotor Resistance Starter

Slip ring motors are always started with a resistance in series with each rotor phase in this starter, full line voltage is applied across starter windings. Full value additional resistance is added in series with each phase of the rotor windings. These resistances are immersed in oil. As the rotor starts up, the handle is rotated slowly until the entire resistance is out. The motor then runs normally. At this stage, rotor windings are short- circuited eliminating the resistance. Now the motor runs like a squirrel cage motor.



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